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Mera Hisaab

Apni Dukaan, Apna Hisaab. — A digital hisaab-kitaab partner for every local shop.

Project & User Guide | Version: Phases 1–5 complete | 69 automated tests passing

1. What is Mera Hisaab?

A simple, mobile-first business-management (ERP) web app for Indian local shops. One shopkeeper can manage products, sales/billing, purchases, stock, customers, udhaar (credit), suppliers, and profit — all from one place. It is a **multi-tenant SaaS**: one application safely serves thousands of independent shops, and each shop's data is completely isolated from every other shop.

Everything is **dynamic and database-driven** — no hardcoded shops, products, prices, categories, payment methods, or dashboard numbers. Every figure you see is calculated live from the database.

2. Technology

Layer	Technology
Backend	Python + Django 5.2 (Django ORM)
Database	PostgreSQL (production target); SQLite for quick local development
Frontend	Django templates + Bootstrap 5 + HTMX + a little vanilla JavaScript
PDF invoices	xhtml2pdf
Fonts/assets	Self-hosted (Inter, Noto Sans Devanagari, Bootstrap, HTMX) — no external CDNs

3. How to run the app

1. Open a terminal in the project folder `C:\Users\Kanishak\Desktop\mera_hisaab`.
2. Create an admin login (one time): `.\venv\Scripts\python.exe manage.py createsuperuser`
3. Start the server: `.\venv\Scripts\python.exe manage.py runserver`
4. Open `http://127.0.0.1:8000/` in your browser.

Database: The app currently uses a SQLite file, `db.sqlite3`, in the project folder (created automatically). To switch to PostgreSQL, set `USE_SQLITE=0` and the `POSTGRES_*` values in the `.env` file, then run `migrate`. View data anytime via the Admin panel, or open `db.sqlite3` in **DBeaver** (New Connection → SQLite → pick the file).

4. First-time flow

Register → Create your business (onboarding) → Pick business type → Dashboard → Add products → Add customers → Purchase stock → Make sales. Stock, udhaar, and profit all update automatically.

5. Feature guide (how to use each part)

Login & business

Log in with your **email** and password. You can own or belong to several shops; switch between them from the **business dropdown** in the top bar. Toggle **dark mode** with the moon icon.

Dashboard

Live cards: Today's Sales, Purchases, Expenses, Profit, Customers, Pending Udhaar, Products, Low Stock. Use the **Period** filter (Today / Week / Month / Custom, etc.) to change the range. All numbers are real, from your data.

Products & Categories

1. **Products** → **Add Product**: name, SKU, barcode, prices (purchase & selling), tax %, unit, minimum stock, image, and an **opening stock**.
2. Extra fields appear automatically based on your **business type** (e.g. a Mobile shop sees IMEI/Model; Garments sees Size/Color).
3. **Categories**: group products (e.g. Rice, Snacks). Products and categories can be deactivated (soft-delete), never lost.

Inventory

An overview of stock on hand, minimum level, and stock value. The **Low stock** button filters items at/below their minimum so you know what to reorder. Use **Adjust** to correct a quantity (breakage, counting) — it records an adjustment in the item's history. Normally stock moves by itself: purchases increase it, sales decrease it.

Customers & Udhaar (Khata)

1. **Customers** → **Add Customer** (name, phone, credit limit, notes).
2. Each customer has a **ledger**: a positive balance means they owe you (udhaar).
3. **Add udhaar** increases their balance; **Record payment** reduces it. Over-payment becomes an advance.
4. **Udhaar** page: all customers who owe money, most-owed first, with total outstanding and one-click Collect.

POS / Sales (billing)

1. Open **POS / Sales** (or the Sell tab on mobile). Search or scan a product and tap to add it to the cart.
2. Adjust quantity and discount. Choose a **customer** (needed if any amount stays on udhaar) and a **payment method**.
3. Click **Complete Sale**. The app then, in one safe step: creates the invoice, reduces stock, records the payment, puts any unpaid balance on the customer's udhaar, and calculates profit.
4. On the invoice you can **Print**, download **PDF**, or **Share on WhatsApp** (a link with the invoice summary and a public view page, ending with “Powered by Mera Hisaab”).

> All money is calculated on the server (never trusted from the browser), using exact decimals. Invoice numbers like `MH-2026-000001` are generated per shop, in order, with no duplicates.

Suppliers & Purchases

1. **Suppliers** → **Add Supplier**. Each supplier has a payable ledger (what you owe them).
2. **Purchases** → **New Purchase**: add products with quantity and cost, choose the supplier and how much you paid now.
3. On save: **stock increases automatically**, the product's cost is updated (for accurate profit), and any unpaid amount becomes **supplier payable**. Pay it later from the supplier's page.

Payment methods

Each shop gets an editable set (Cash, UPI, Card, Bank Transfer, Credit/Udhaar, Other). Choosing Credit/Udhaar sends the amount to the customer's khata instead of counting it as received. These are stored in the database, not hardcoded.

Super Admin panel

At `/admin/` (needs a superuser account). This is the platform-owner view: manage shops, users, business types & their dynamic fields, and inspect all records. Money fields are read-only there on purpose — they change only through the app's logic.

6. Security & correctness

- **Tenant isolation**: every record belongs to a business; the active shop comes from your logged-in session, never from a browser-supplied id. Automated tests confirm one shop cannot see or touch another shop's data.
- **Money**: stored as exact decimals (never floating point); all totals, tax, discounts and balances computed on the backend.
- **Consistency**: sales, purchases, and ledger updates run inside atomic, row-locked transactions — they either complete fully or roll back; stock and balances never drift.
- CSRF protection, secure password hashing, and Django's ORM (no raw SQL) are used throughout.

7. What is built vs. planned

Phase	Scope	Status
1	Auth, business onboarding, multi-tenancy, dashboard, admin	**Done**
2	Products, categories, dynamic fields, inventory ledger, low stock	**Done**
3	Customers, udhaar ledger, payments	**Done**
4	POS/Sales, payments, invoices (print/PDF/WhatsApp)	**Done**
5	Suppliers, purchases, auto stock-in, payables	**Done**
6	Expenses + true net Profit & Loss (Sales – COGS – Expenses)	**Next**
7	Reports + full GST + CSV/PDF export + global search	**Planned**
8	Roles & permissions (RBAC), audit log, notifications	**Planned**
9	Subscription plans (SaaS monetization)	**Planned**
10	PWA/offline, REST API, Docker, deployment hardening	**Planned**

8. Known limitations (today)

- No sale/purchase return or cancel yet (documents are immutable once posted).
- GST is a single tax rate per product; full CGST/SGST/IGST + HSN is Phase 7.
- WhatsApp shares a link (not an attached PDF file) — a platform limitation without the WhatsApp Business API.
- Roles are shop-level (owner/staff) for now; fine-grained permissions come in Phase 8.
- Expenses and true net profit are Phase 6 (dashboard Profit currently shows gross profit from sales).